Prajwal Niraula

ASTROPHYSICIST / EXOPLANET ATMOSPHERES / COMPUTATIONAL PHYSICS

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Research Profile

Astrophysicist developing physically grounded models and high-performance computational tools to interpret exoplanet spectra. Research combines molecular opacity and pressure-broadening calculations, Bayesian atmospheric retrievals, stellar-contamination models, and multi-instrument observations from HST, JWST, Kepler/K2, and TESS.

Research Experience

Postdoctoral Associate Jun 2023–Present
Massachusetts Institute of Technology

- Lead studies of stellar contamination in exoplanet transmission spectroscopy, integrating HST/STIS, HST/WFC3, Kepler, and TESS observations to strengthen atmospheric inferences.
- Develop Bayesian retrieval and forward-modeling frameworks that combine PHOENIX stellar spectra, flux-calibrated HST spectroscopy, broadband spectral energy distributions, and rotational photometry.
- Calculate *ab initio* molecular pressure-broadening properties using quantum dynamics and quantify their impact on JWST atmospheric-abundance retrievals.
- Build reproducible Python and Fortran workflows for spectral synthesis, radiative transfer, high-dimensional inference, and high-performance computing.

Graduate Research Assistant Aug 2018–May 2023
Massachusetts Institute of Technology

- Developed algorithms supporting exoplanet discovery with the ground-based SPECULOOS survey of ultracool dwarfs.
- Led the discovery of four planets using Kepler/K2 data and contributed to the characterization of additional systems, including GJ 9827.
- Used phase curves and secondary-eclipse analyses to challenge the planetary classification of four previously vetted candidates.
- Built optimized absorption cross-section and radiative-transfer software covering billions of spectral transitions; used Bayesian inference and high-performance computing to investigate opacity-driven uncertainties in JWST observations.

Resident Astronomer Sep 2015–Jul 2016
Atacama Cosmology Telescope, Chile

- Supported high-altitude telescope operations, including instrument inspection, troubleshooting, maintenance, and observing coordination.
- Collaborated on science and engineering data acquisition, diagnostic testing, and performance monitoring, drawing on experience fabricating and machining experimental hardware.

Education

Ph.D. in Earth, Atmospheric and Planetary Sciences 2018–2023
Massachusetts Institute of Technology, Cambridge, MA
Thesis: *Towards Robust Inferences about Exoplanets and Their Atmospheres*. Advisor: Julien de Wit.

Master's in Astronomy with Planetary Science 2016–2018
Wesleyan University, Middletown, CT
Thesis: *Planet Hunting and Characterization with K2*. Advisor: Seth Redfield.

B.S. in Physics and Mathematics 2011–2015
Saint Peter's University, Jersey City, NJ
Co-valedictorian. Thesis: *Experimentation and Modeling of Self-Organization in Cathode Boundary Layer Discharge in Noble Gas*. Advisor: WeiDong Zhu.

Selected Publications

First-author and collaborative studies in exoplanet atmospheres, spectroscopy, and orbital analysis.

1. **P. Niraula**, L. Wiesenfeld et al. *Comprehensive Ab Initio Quantum Computations of CO_2–H_2 and CO_2–He Collisional Properties*. *The Astrophysical Journal*, 2026. doi:10.3847/1538-4357/ae83a3.
2. **P. Niraula**, B. V. Rackham et al. *Unlocking HST's Stellar Treasure Trove: Stellar Activity Minima for HAT-P-11 Offer Prime Windows for Transmission Spectroscopy*. *The Astronomical Journal*, 2026. doi:10.3847/1538-3881/ae83b4.
3. **P. Niraula**, J. de Wit et al. *The Detection-versus-retrieval Challenge: Titan as an Exoplanet*. *The Astrophysical Journal Letters*, 2025. doi:10.3847/2041-8213/ae27c2.
4. **P. Niraula**, J. de Wit et al. *Origin and Extent of the Opacity Challenge for Atmospheric Retrievals of WASP-39 b*. *The Astrophysical Journal Letters*, 2023. doi:10.3847/2041-8213/acd6f8.
5. **P. Niraula**, J. de Wit et al. *The Impending Opacity Challenge in Exoplanet Atmospheric Characterization*. *Nature Astronomy*, 2022. doi:10.1038/s41550-022-01773-1.
6. **P. Niraula**, A. Shporer, I. Wong, and J. de Wit. *Revisiting Kepler Transiting Systems: Unvetting Planets and Constraining Relationships among Harmonics in Phase Curves*. *The Astronomical Journal*, 2022. doi:10.3847/1538-3881/ac4f64.
7. **P. Niraula**, J. de Wit et al. *π Earth: A 3.14 Day Earth-sized Planet from K2's Kitchen Served Warm by the SPECULOOS Team*. *The Astronomical Journal*, 2020. doi:10.3847/1538-3881/aba95f.
8. **P. Niraula**, S. Redfield, F. Dai et al. *Three Small Super-Earths Transiting the Nearby Star GJ 9827*. *The Astronomical Journal*, 2017. doi:10.3847/1538-3881/aa957c.

Selected Observing Programs

Co-Investigator on the following observing programs.

- **HST GO 16875 (2022)** — *Reconnaissance Transmission Spectroscopy of the Best Temperate Mini-Neptune for Atmospheric Characterisation around TOI-4336 b*. PI: Georgina Dransfield.
- **HST GO 16446 (2022)** — *Two Birds, One Stone: Confirming Two Transiting Moon-sized Exoplanets at 6.9 pc*. PI: Benjamin Rackham.
- **Kepler/K2 GO 19038 (2019)** — *Monitoring Targets Within 100 Parsecs*. PI: Seth Redfield.
- **Spitzer GO 15434 (2018)** — *Three Transits for the Price of One: Super-Earth Transits of the Nearest Planetary System Discovered by Kepler/K2*. PI: Seth Redfield.
- **HST GO 16425 (2017)** — *Searching for Atmospheric Loss Signatures in the Newly Discovered Super-Earth GJ 9827 b*. PI: Seth Redfield.

Selected Presentations

The Impending Opacity Challenge in Exoplanet Atmospheric Characterization. International Symposium on Molecular Spectroscopy 2025

A Scalable Solution to the Opacity Challenge. 245th American Astronomical Society Meeting 2025

The Impending Opacity Challenge in Exoplanet Atmospheric Characterization. International Symposium on Molecular Spectroscopy 2023

The Impact of Opacity on the Interpretation of Transmission Spectra. Center for Astrophysics | Harvard & Smithsonian, Exoplanet Pizza Talk 2023

Invalidating Planets Using Phase Curves. MIT EAPS, PICS Talk 2022

Technical Expertise

Methods: Bayesian inference, forward modeling, radiative transfer, stellar spectral modeling, quantum molecular dynamics, time-series analysis.

Computing: Python, Fortran, high-performance computing, scientific software development, reproducible data analysis.

Observations: JWST, HST/STIS and WFC3, Kepler/K2, TESS, SPECULOOS.